

SUPPLEMENTARY MATERIAL: A MULTIMODAL INTRINSICS-GUIDED THERMAL-AWARE FRAMEWORK FOR RGB LOW-LIGHT IMAGE ENHANCEMENT

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1. DETAILED NETWORK ARCHITECTURE

The proposed framework employs a dual-encoder U-Net architecture. During our experimentation, we found out that utilizing Group Normalization (GN) instead of Batch Normalization throughout the network was helpful in mitigating the training instability and achieving overall better results.

1.1. Fundamental Building Blocks

In this section we detail the hyperparameters of the core computational units:

- **ConvBlock:** The standard processing unit consists of a 3×3 Convolution (with padding 1, bias=False), followed by Group Normalization ($N_{groups} = 8$) and a LeakyReLU activation (slope $\alpha = 0.2$).
- **ResBlock:** Utilized in the refinement stage, it consists of a ConvBlock followed by a second 3×3 Convolution and Group Normalization. A residual connection adds the input feature map to the output before a final LeakyReLU activation ($\alpha = 0.2$).

1.2. Encoder-Decoder Configuration

The backbone is a symmetric 4-level U-Net. To maintain a lightweight architecture, we set the initial network width to $C = 16$ channels for both inputs.

- **Dual Encoders:** We utilize two independent encoders. Downsampling is performed via strided convolutions ($k = 3, s = 2, p = 1$), doubling the channels at each stage ($16 \rightarrow 32 \rightarrow 64 \rightarrow 128$).
- **Bottleneck:** The bottleneck concatenates the deepest features from both modalities ($128 + 128$) and processes them via a ConvBlock with 256 channels, preserving the spatial resolution of $H/8 \times W/8$.
- **Decoder & Fusion:** Upsampling is performed via bi-linear interpolation followed by a concatenation with the gated skip connections. The skip connections are fused via a 1×1 convolution that reduces the concatenated channels ($C_l + C_l$) back to C_l before entering the Attention Gate.

- **Prediction Heads:** The decoder branches into two heads for Reflectance (R) and Inverse Shading (D). Each head consists of: 3×3 Conv $\rightarrow$ LeakyReLU $\rightarrow 3 \times 3$ Conv $\rightarrow$ Sigmoid, mapping the 16-channel feature map to 3 channels (for R) or 1 channel (for D).

1.3. Refinement Module

The refinement network is designed to recover details lost during the coarse prediction. It accepts a 4-channel input obtained by concatenating $\hat{I}_{coarse}$ (3 ch) and the original thermal input T (1 ch). First, a ConvBlock maps the 4-channel input to a hidden dimension of 32 channels; then a sequence of 4 ResBlocks processes the features at full resolution ($H \times W$) to capture fine structural dependencies; finally a 1×1 convolution projects the features to 3 channels, producing the residual map $\mathcal{E}$. The final output is obtained as $\hat{I}_{out} = \sigma(\hat{I}_{coarse} + \mathcal{E})$, where σ is the Sigmoid function.

2. LOSS COMPONENTS DESCRIPTION

In this section, we present more in detail the implementation of the loss functions used to supervise the network. For all formulations, we use Y to denote the ground truth and $\hat{Y}$ the prediction; N is the total number of pixels.

Pixel Reconstruction Loss ($\mathcal{L}_{pix}$). To compute the photometric error, we employ the Smooth ℓ_1 loss, which is less sensitive to outliers than ℓ_2 . It is defined as:

$$\mathcal{L}_{pix}(\hat{Y}, Y) = \frac{1}{N} \sum_p \rho_{\delta}(\hat{Y}_p - Y_p),$$

$$\rho_{\delta}(d) = \begin{cases} 0.5 \cdot d^2 / \delta, & \text{if } |d| < \delta, \\ |d| - 0.5 \cdot \delta, & \text{otherwise,} \end{cases} \quad (1)$$

where δ is set to 0.01. This loss is always applied with weight $\lambda_p = 1.0$.

Edge Loss ($\mathcal{L}_{edge}$). To preserve structural sharpness, we use a composite loss constraining both gradient magnitude and orientation. Let ∇ denote the spatial gradient operator and $M(\cdot) = \|\nabla \cdot\|_2$ its magnitude. The function is defined as:

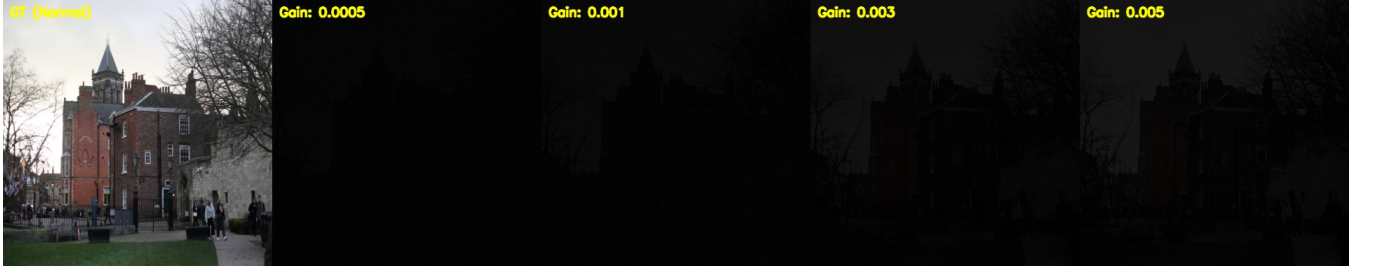


Fig. 1. Low light simulation from HDRT ground truth normal-exposed image using different values for the exposure gain. Best if viewed in digital format.

$$\begin{aligned}\mathcal{L}_{edge}(\hat{Y}, Y) &= \mathcal{L}_{mag} + 0.5 \cdot \mathcal{L}_{dir} \\ \mathcal{L}_{mag} &= \|M(\hat{Y}) - M(Y)\|_1 \\ \mathcal{L}_{dir} &= \frac{1}{N} \sum_p \mathbf{1}_{\{M(Y)_p > \tau\}} \cdot \left(1 - \frac{\nabla \hat{Y}_p \cdot \nabla Y_p}{M(\hat{Y})_p \cdot M(Y)_p + \varepsilon}\right)\end{aligned}\quad (2)$$

where $\mathbf{1}_{\{\cdot\}}$ is a function that aims to mask flat regions (threshold $\tau = 0.05$) to ensure numerical stability and p is any pixel. The weight here is always set to $\lambda_e = 0.8$.

Color Loss ($\mathcal{L}_{lab}$). To decouple chromaticity from luminance, we first convert RGB images to the CIELAB space and minimize the error specifically on the chrominance channels (a, b):

$$\mathcal{L}_{lab}(\hat{Y}, Y) = \|\hat{Y}_a - Y_a\|_1 + \|\hat{Y}_b - Y_b\|_1 \quad (3)$$

We explicitly exclude the luminance (L) channel to enforce color consistency independent of brightness. We set the weight $\lambda_c = 1.0$.

Perceptual Loss ($\mathcal{L}_{perc}$). To align results with human perception and preserve semantic consistency, we compute the distance between feature maps extracted from a pre-trained VGG-16 network [1]. We extract maps ϕ_j from four distinct depths: shallow layers (relu1_2, relu2_2) capturing fine textures, and deeper layers (relu3_3, relu4_3) capturing structural semantics. The loss is defined as:

$$\mathcal{L}_{perc}(\hat{Y}, Y) = \sum_{j=1}^4 w_j \cdot \|\phi_j(N(\hat{Y})) - \phi_j(N(Y))\|_1 \quad (4)$$

where $N(\cdot)$ represents the standard ImageNet mean-variance normalization required by the pre-trained backbone. The weights are set to $w = \{1/32, 1/16, 1/8, 1/4\}$ respectively, assigning higher importance to higher-level semantic features. This component is weighted in the total loss with $\lambda_\phi = 0.2$.

Total Objective. The final objective function combines these components for the reflectance (R), shading (S), physical con-

sistency (phy), and refinement (ref) stages:

$$\mathcal{L}_{total} = \alpha(\mathcal{L}_R + \mathcal{L}_S) + \beta \mathcal{L}_{phy} + \gamma \mathcal{L}_{ref}, \quad (5)$$

where we set $\alpha = 1, \beta = 0.5, \gamma = 1$.

3. PHYSICS-BASED LOW-LIGHT SIMULATION

In this section we extend the description of the adaptive exposure scaling and noise injection part in the low-light simulation.

Adaptive Scaling. During training, instead of a fixed gain, we determine the exposure factor κ dynamically to target specific darkness levels regardless of the initial scene brightness of the image $I_{lin} = I_{gt}^\gamma$. We define a target mean intensity t_{mean} sampled uniformly from $[0.0005, 0.005]$. The gain is computed as:

$$\kappa = t_{mean} / (\mu(I_{lin}) + \varepsilon) \quad (6)$$

where $\mu(\cdot)$ represents the mean operator. Figure 1 shows several low-light simulated version of the same image with different values of κ .

Noise Model. Following the signal-dependent nature of photon arrival, we model the degradation process in the linear domain as:

$$I_{noisy} = \underbrace{\kappa \cdot I_{lin}}_{Signal} + \underbrace{\mathcal{N}_s(0, \sigma_s^2) \cdot \sqrt{\kappa \cdot I_{lin}}}_{Shot Noise \, n_{shot}} + \underbrace{\mathcal{N}_r(0, \sigma_r^2)}_{Read Noise \, n_{read}} \quad (7)$$

where $\mathcal{N}$ denotes the standard Normal distribution. To ensure robustness across different noise levels, we sample $\sigma_s \in [0.001, 0.01]$ and $\sigma_r \in [0.00005, 0.0002]$.

4. ADDITIONAL VISUAL COMPARISON

In Figures 3 and 4 are shown additional results and comparisons between the proposed solution and state-of-the-art methods on LLVIP and V-TIEE dataset. Note that for LLVIP images we used the test set of 70 samples that Jha et al. [2]

made available from their official GitHub page ¹. Regarding V-TIEE images, due to the lack of official documentation detailing the pairing of low-light and well-lit images (ground truths), we performed a manual selection and pairing between the different low-exposure and high-exposure versions within the dataset. This naturally led to the formation of a custom test set. Furthermore, since some well-lit images intended to serve as ground truth presented clearly perceptible artifacts (e.g., under-exposure, noise, incorrect white balance), we also selected images from the dataset that had balanced and clean ground truths, reducing the full set to a subset of 28. We report in Figure 2 the 28 V-TIEE normal light images used in the test as ground truths. While this filtering was strictly intended to avoid evaluating on corrupted targets, we acknowledge that manually selecting a subset may induce bias, and we explicitly highlight these artifact issues as a dataset limitation.

5. REFERENCES

- [1] Karen Simonyan and Andrew Zisserman, “Very deep convolutional networks for large-scale image recognition,” 2015.
- [2] Raman Jha, Adithya Lenka, Mani Ramanagopal, Aswin Sankaranarayanan, and Kaushik Mitra, “Rt-x net: Rgb-thermal cross attention network for low-light image enhancement,” 2025.
- [3] Yifan Jiang, Xinyu Gong, Ding Liu, Yu Cheng, Chen Fang, Xiaohui Shen, Jianchao Yang, Pan Zhou, and Zhangyang Wang, “Enlightengan: Deep light enhancement without paired supervision,” 2021.
- [4] Yufei Wang, Renjie Wan, Wenhan Yang, Haoliang Li, Lap-Pui Chau, and Alex C Kot, “Low-light image enhancement with normalizing flow,” *arXiv preprint arXiv:2109.05923*, 2021.
- [5] Yuanhao Cai, Hao Bian, Jing Lin, Haoqian Wang, Radu Timofte, and Yulun Zhang, “Retinexformer: One-stage retinex-based transformer for low-light image enhancement,” 2023.

¹<https://github.com/jhakrraman/rt-xnet>



Fig. 2. The 28 V-TIEE images manually selected and used for testing.

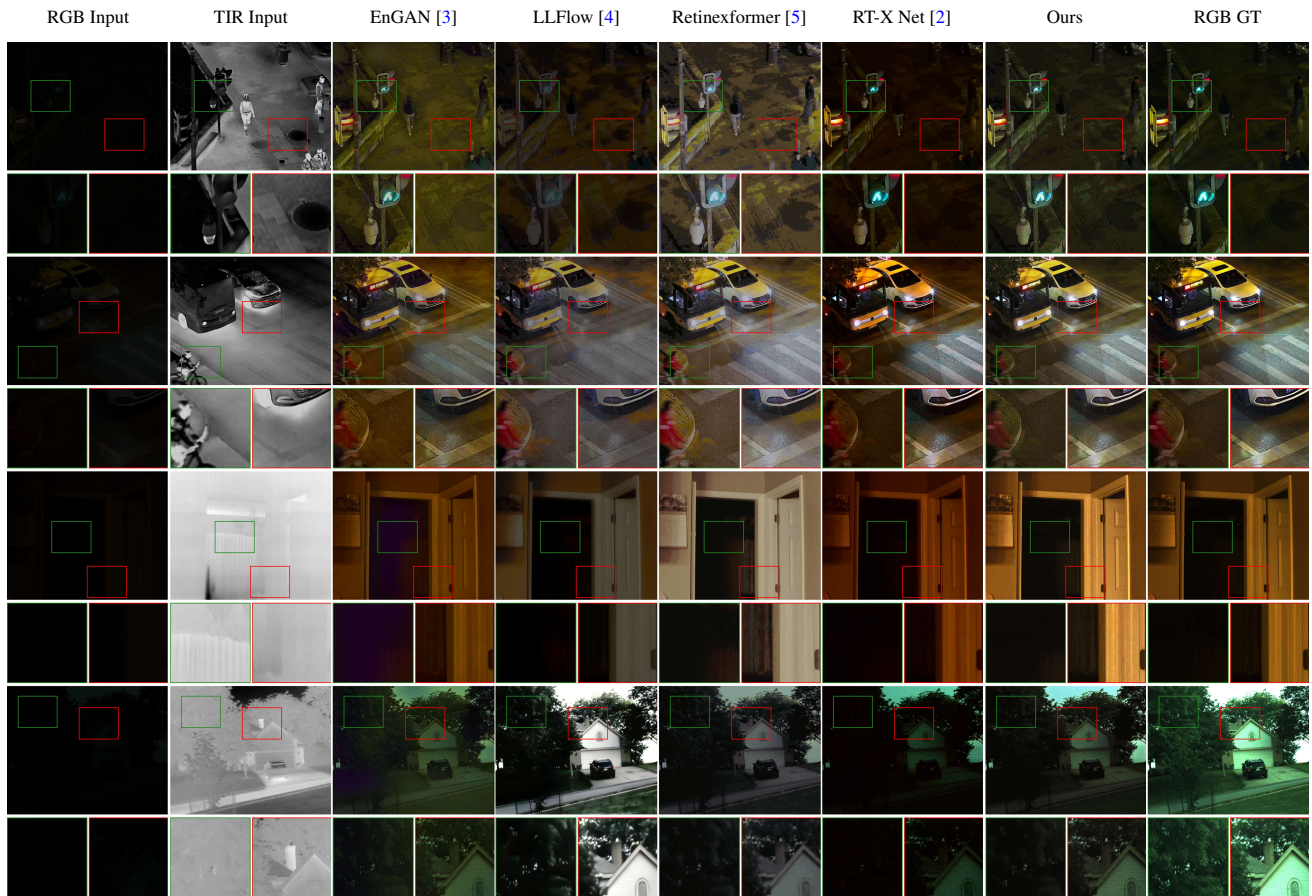


Fig. 3. Visual comparison example. The first two rows show low-light samples from the LLVIP dataset, while the last two rows are from V-TIEE. Each column compares the analyzed method against the ground truth.

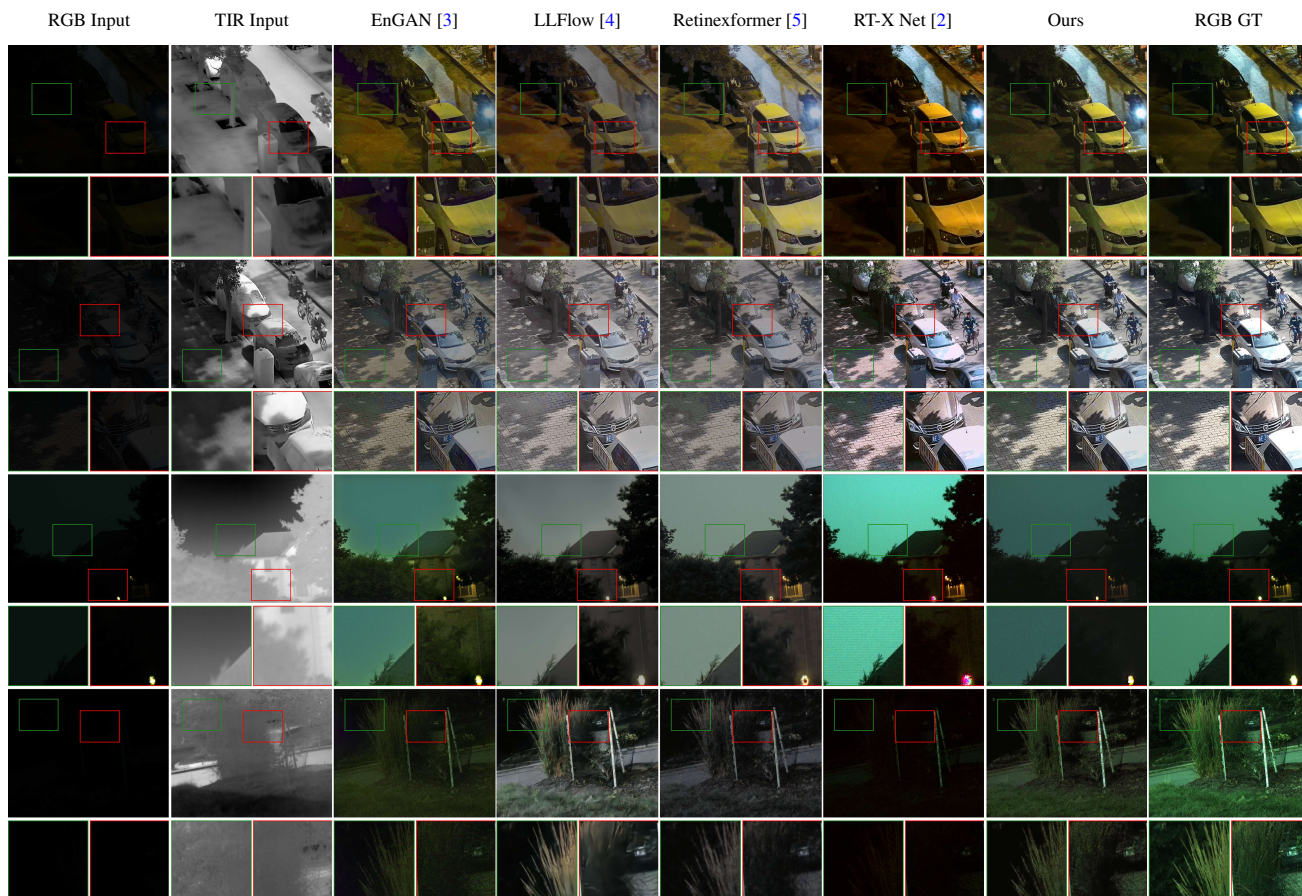


Fig. 4. Visual comparison example. The first two rows show low-light samples from the LLVIP dataset, while the last two rows are from V-TIEE. Each column compares the analyzed method against the ground truth.